

Regime-Dependent Variance Risk Premium

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1 Why Does VRP Predict Returns Only in High-Volatility Regimes?

1.1 The Empirical Puzzle

Our preliminary time-series analysis reveals a striking pattern: the variance risk premium’s ability to predict future 50ETF returns is concentrated in high-volatility regimes. The Z-score normalized VRP achieves an R^2 of 6.14% ($t = 3.82$) during panic periods, compared to near-zero predictability ($R^2 < 0.15\%$) in low and medium volatility regimes. This section synthesizes the theoretical literature to explain this regime-dependent predictability.

1.2 Channel 1: Time-Varying Risk Aversion

Bollerslev, Tauchen, and Zhou (2009) and Drechsler and Yaron (2011) interpret VRP as a proxy for time-varying risk aversion. In their framework, periods of high VRP coincide with periods when investors become more fearful about future uncertainty—not just about volatility levels, but about volatility-of-volatility.

Why regime-dependence? Risk aversion itself is state-dependent. During calm periods, investors are complacent and small VRP fluctuations reflect noise rather than genuine shifts in risk aversion. During panic periods, investors become highly risk-averse and VRP surges as they scramble to hedge. The covariance between VRP and consumption risk becomes strongly negative, making VRP a powerful predictor of mean reversion.

Bekaert and Hoerova (2014) decompose the VIX into risk-aversion and uncertainty components. They find that the risk-aversion component exhibits sharp spikes during crises and is a much stronger predictor of returns than the uncertainty component.

Cross-sectional implication: Stocks with high VRP betas amplify portfolio risk during high-aversion regimes and should command a **positive risk premium**. Alternatively, stocks that hedge VRP shocks (negative betas) should earn lower returns.

1.3 Channel 2: Jump Risk and Tail Events

Todorov (2010) and Eraker and Wang (2015) demonstrate that VRP primarily compensates for **jump risk**—the fear of sudden, catastrophic crashes. They decompose VRP into continuous volatility and jump components, showing that the jump component explains the majority of the variance risk premium.

Why regime-dependence? Jump risk is episodic. During calm periods, jump intensity is low and markets trade in a diffusion-dominated regime. The gap between implied and realized volatility is small and noisy. During panic periods, jump intensity surges and investors overpay for crash protection. When the feared jump does not materialize, volatility mean-reverts, generating predictable returns.

Santa-Clara and Yan (2010) provide empirical support by separating VRP into jump and diffusive components using high-frequency data. They find that the jump component is the primary driver of return predictability, particularly during crises. The diffusive component has near-zero predictive power.

Cross-sectional implication: Stocks with high VRP betas are “short tail risk”—they crash hardest during tail events. Crash-averse investors demand a **positive risk premium** to hold these stocks. This channel unambiguously predicts a positive VRP premium in the cross-section.

1.4 Channel 3: Intermediary Capital Constraints

He and Krishnamurthy (2013) and Chen, Joslin, and Ni (2019) argue that VRP spikes when financial intermediaries hit capital constraints. Intermediaries are the primary sellers of volatility insurance (they write options). When capital is scarce, they demand higher compensation for bearing volatility risk, causing implied volatility to surge relative to realized volatility.

Why regime-dependence? Intermediary constraints bind only during crises. In normal times, intermediaries have ample capital and provide liquidity at low cost. VRP reflects mostly transaction costs. During stress periods, intermediaries face margin calls and redemptions, withdraw liquidity, and VRP spikes. As they recover capital and re-enter the market, prices normalize and returns exhibit mean reversion.

Haddad and Muir (2021) construct a measure of intermediary distress using broker-dealer balance sheet data. They find that VRP has strong predictive power for equity returns *only when intermediary distress is elevated*. When intermediaries are well-capitalized, VRP is unrelated to future returns.

Cross-sectional implication: Stocks with high VRP betas are expensive for intermediaries to hold and hedge. They perform poorly during deleveraging episodes and require a **positive risk premium** to clear the market. This links VRP exposure to funding liquidity risk.

1.5 Channel 4: Heterogeneous Beliefs

Buraschi, Trojani, and Vedolin (2014) link VRP to disagreement among investors. During crises, divergence in beliefs widens as uncertainty increases. Since options aggregate diverse expectations, increased disagreement inflates implied volatility. VRP becomes elevated, but part of it reflects mispricing rather than a genuine risk premium.

Why regime-dependence? Disagreement is countercyclical. During calm periods, investors broadly agree on the outlook and VRP is small. During panic periods, some believe the crisis will deepen while others expect quick resolution. This disagreement inflates option prices. As new information arrives and disagreement resolves, prices correct, generating return predictability.

Cross-sectional implication: If disagreement represents speculative risk, stocks with high VRP betas may become *overvalued* during high-disagreement periods and subsequently *underperform*. This suggests a **negative "premium"** (actually a reversal from mispricing), distinguishing it from the rational risk channels.

1.6 Why Z-Score Outperforms Raw VRP

Our finding that Z-score VRP achieves 6.14% R^2 compared to 2.15% for raw VRP has a clear theoretical interpretation.

The heteroskedasticity problem: Raw VRP is measured in variance units. During crises, raw VRP can be 10–20 times larger than in normal times simply because baseline volatility is higher. A predictive regression struggles to reconcile these scale differences.

The Z-score solution: The Z-score measures VRP in units of standard deviations from the rolling mean:

$$\text{VRP}_{\text{Zscore}_t} = \frac{\text{VRP}_t - \mu_{\text{VRP},[t-60,t]}}{\sigma_{\text{VRP},[t-60,t]}}$$

This achieves two things: (1) removes scale dependence—a Z-score of +2 in 2018 is directly comparable to +2 in 2022, and (2) captures relative extremity—what matters is whether VRP is *unexpectedly high* relative to recent history.

Adaptive markets interpretation: Financial markets continuously learn and adapt. If VIX stays elevated at 35 for three months, the market adjusts. Raw VRP remains high but no longer predicts returns because the market has adapted. The Z-score captures this adaptation—it reverts toward zero as the moving average catches up. A high Z-score specifically identifies **sudden, unanticipated spikes** in variance risk—the moments when jump risk becomes salient, intermediaries are caught off-guard, and disagreement widens abruptly. These are moments of maximum mispricing and maximum return predictability.

1.7 Summary

Four complementary channels explain why VRP predicts returns only in high-volatility regimes:

1. **Risk Aversion:** Time-varying risk aversion spikes during crises, making VRP a powerful predictor
2. **Jump Risk:** Jump risk is episodic; VRP’s jump component drives predictability during panic periods
3. **Intermediary Constraints:** Intermediary capital constraints bind during crises, causing VRP to spike and predict mean reversion
4. **Disagreement:** Investor disagreement widens during crises, inflating VRP through mispricing

All channels predict regime-dependence, though they differ on mechanism and the expected sign of cross-sectional risk premiums. The Z-score’s superior performance reflects its ability to capture relative extremity—the key dimension all channels agree is most predictive.

2 Cross-Sectional Analysis: The Fama-MacBeth Approach

Having established *why* VRP predicts aggregate returns in a regime-dependent manner, we now turn to whether VRP is priced in the *cross-section* of individual stock returns. Do stocks with different VRP exposures earn different average returns?

2.1 The Asset Pricing Framework

In a multi-factor model, the expected excess return on stock i is:

$$E[R_i - R_f] = \lambda_{VRP} \cdot \beta_{i,VRP} + \lambda_{MKT} \cdot \beta_{i,MKT} + \lambda_{SMB} \cdot \beta_{i,SMB} + \dots \quad (1)$$

where $\beta_{i,VRP}$ is stock i ’s exposure to VRP and λ_{VRP} is the **market price of VRP risk**—the expected excess return per unit of VRP beta. Our goal is to estimate λ_{VRP} and test whether it differs significantly from zero.

2.2 The Fama-MacBeth Two-Step Procedure

The methodology consists of three stages:

2.2.1 Step 0: Estimate Factor Betas (Time-Series Regressions)

For each stock i , we run a time-series regression using a 60-month rolling window:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{i,VRP} \cdot VRP_t + \beta_{i,MKT} \cdot MKT_t + \beta_{i,SMB} \cdot SMB_t + \beta_{i,HML} \cdot HML_t + \epsilon_{i,t} \quad (2)$$

This yields estimates $\hat{\beta}_{i,VRP}(t)$, $\hat{\beta}_{i,MKT}(t)$, etc. for each stock at each time t .

2.2.2 Step 1: Cross-Sectional Regressions (Estimate Risk Premia)

At each month t , we run a *cross-sectional* regression across all stocks:

$$R_{i,t+1} - R_{f,t+1} = \lambda_{0,t} + \lambda_{VRP,t} \cdot \hat{\beta}_{i,VRP}(t) + \lambda_{MKT,t} \cdot \hat{\beta}_{i,MKT}(t) + \dots + \eta_{i,t+1} \quad (3)$$

This regression uses realized returns from t to $t + 1$ and betas estimated through time t (no look-ahead bias). The coefficient $\hat{\lambda}_{VRP,t}$ is the estimated price of VRP risk in month t . We repeat this for every month, obtaining a time series: $\{\hat{\lambda}_{VRP,1}, \hat{\lambda}_{VRP,2}, \dots, \hat{\lambda}_{VRP,T}\}$.

2.2.3 Step 2: Time-Series Averaging (Test Significance)

We compute the average risk premium:

$$\bar{\lambda}_{VRP} = \frac{1}{T} \sum_{t=1}^T \hat{\lambda}_{VRP,t} \quad (4)$$

and test whether it is significantly different from zero using:

$$t_{\lambda_{VRP}} = \frac{\bar{\lambda}_{VRP}}{\text{SE}(\hat{\lambda}_{VRP})} \quad (5)$$

where SE is the Newey-West standard error accounting for serial correlation in the λ 's.

2.3 Why Fama-MacBeth?

The Fama-MacBeth procedure has three key advantages:

1. Handles cross-sectional correlation: By running separate regressions each month, we collapse the cross-sectional dimension. Instead of $N \times T$ residuals (which are correlated across stocks), we have only T estimates of $\lambda_{VRP,t}$. This naturally accounts for cross-sectional correlation.

2. Addresses errors-in-variables: The betas $\hat{\beta}_{i,VRP}$ are estimated (Step 0), introducing measurement error. Shanken (1992) provides a correction:

$$\text{SE}_{\text{Shanken}}(\hat{\lambda}_{VRP}) = \text{SE}(\hat{\lambda}_{VRP}) \times \sqrt{1 + \frac{\bar{\lambda}_{VRP}^2}{\sigma^2(VRP_t)}}$$

This inflates standard errors to account for beta estimation uncertainty.

3. Allows time-varying risk premia: By estimating $\lambda_{VRP,t}$ separately each month, we can analyze *when* the premium is high or low, linking it to market regimes.